

CPE 301 Final Project Part 2

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1 Introduction

The main problem we are trying to solve is to improve on the canes that blind people use. Currently, most canes are used by tapping or rolling around. Our system improves on this by adding a sensor that looks ahead, detecting walls and other obstacles further than the extended range of the cane. This works for most blind people that aren't deaf that still have trouble navigating through the world. This is a reasonable solution as it improves on an already existing solution without making things too complicated.

2 Proposed System Overview

The main feature of our system will use an ultrasonic sensor and a buzzer. Using the sensor, we should be able to detect up to five feet in front of the extended range of the cane, sending a signal when detecting something in front of it. The buzzer is then used to relay how close the obstacle is, beeping faster the closer the sensor is to the object. There will also be a switch that can change the mode. An indoor and outdoor mode will be implemented, changing the sensitivity of the ultrasonic sensor for further or closer environments.

2.1 System Schematic Diagram

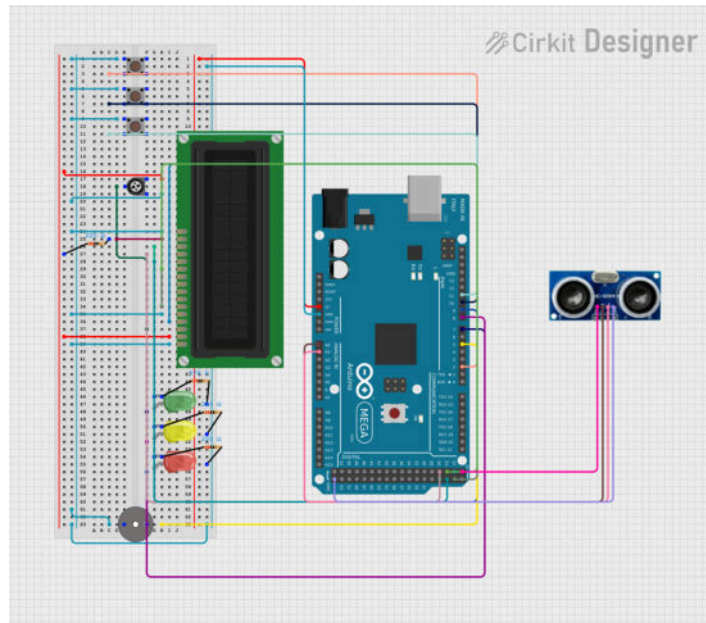


Figure 1: This is the block diagram that describes the system component's actions and how they relate/affect each component in order to achieve the design's purpose

2.2 Component List

Microcontroller: ATmega2560

Ultrasonic Sensor: HC-SR04 Ultrasonic Distance Sensor

LEDs: Standard 5mm LEDs x3 (green, red, yellow)

Audio: Active piezo buzzer (5V)

User Input: Push buttons x3

Display: LCD Display (1602 LDC Display)

Resistors: 220ohm x3, 10k ohm x1

Power: 9V battery

3 Expected System Behavior (High-Level)

When the system is turned on, the ultrasonic sensor will be constantly reading information. Depending on the mode it is on, the sensitivity of the ultrasonic sensor will be adjusted to compensate if the cane is being used outdoors for further objects, or indoors where objects are more likely to be closer. When an object ahead is detected, the buzzer will go off in intervals, activating in a faster interval if the object is closer, and a slower interval when the object is further. When the system is off, nothing is going on with the sensors.

3.1 Module Design

Our only sensor reading module is the ultrasonic sensor we will be using. The ultrasonic sensor will detect obstacles in front of the cane by emitting sound waves at a high frequency, measuring the time it takes for the echo to return. Our control logic module will control this sensor, sending out the ultrasonic signal at certain intervals, sending in an input once the sensor detects something. This will also measure the distance of the object, by measuring the pulse using a timer and converting it to a distance value. Based on this distance, it will determine if the object is within a certain range, which will make a buzzer activate once an object is too close.

Our control module that we plan to use is the ATmega2560. This will run our code as well as read inputs from the ultrasonic sensor, as well as send outputs to our buzzer and display device. The measurements taken from the ultrasonic sensor will be converted into a distance value. Our display device will indicate one of the four states, whether it is off, idle, active, or currently having an error. It will also display the distance from an object, or display if an object is not close enough. As for the buzzer, it will only activate if it is close enough to an object. We will also have three LEDs, a green one that indicates if it is on or idle, and a red one that indicates if it is having an error.

We are using a display module, as well as a buzzer as an auditory display. The first module is an LCD display, displaying various things. One thing that it will display is the state that the system is currently in, whether it is on, idle, off, or currently having an error. It will also show if the cane is not in range of an object, or the distance between an object and the cane when it is in display. As for the buzzer, it will only alert the user when the cane is in range, beeping faster the closer it is in range of an obstacle.

The communication module we will be using is the built in UART (Universal Asynchronous Receiver/Transmitter). This will let us send and receive data by connecting the USB from the ATmega2560 to a computer. It will let us monitor and debug the device. It can send information like distance values and system status messages. It will also help identify errors such as sensor readings. This serial communication will set a baud rate to ensure reliable data transfer.

3.2 Control Logic (State Diagram or Flowchart)

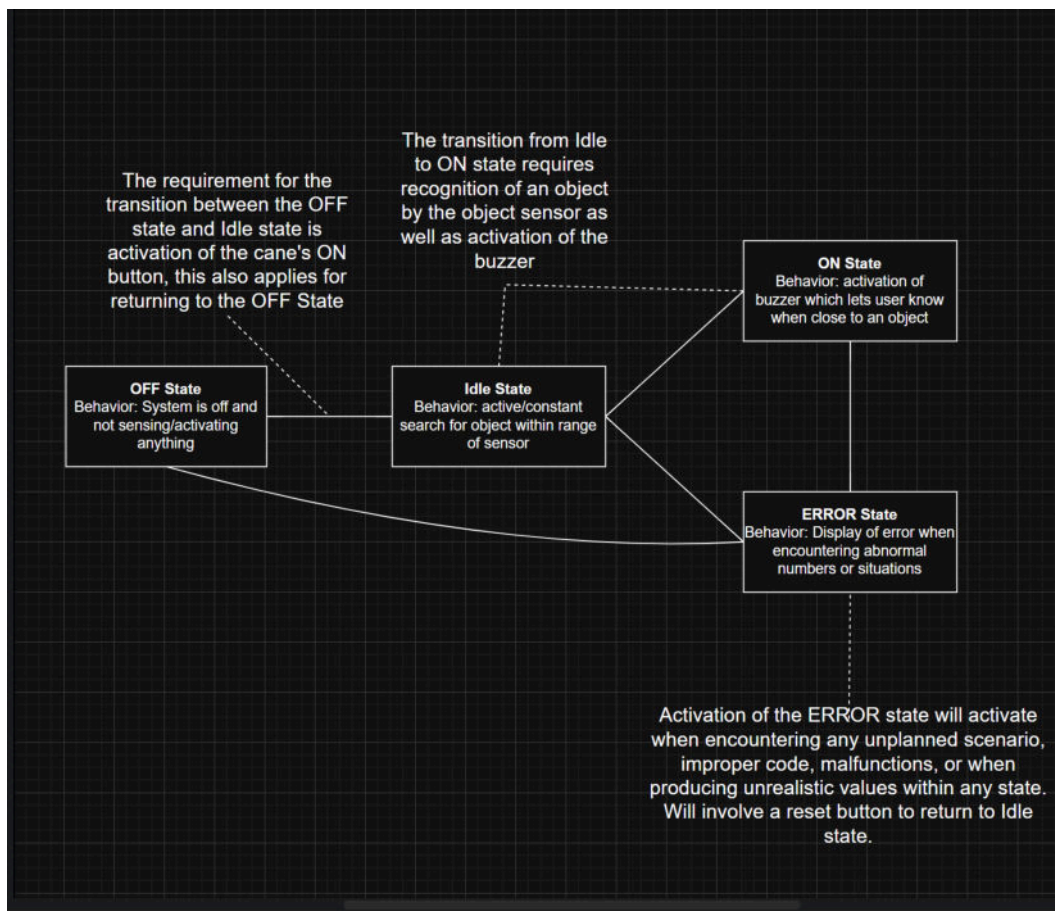
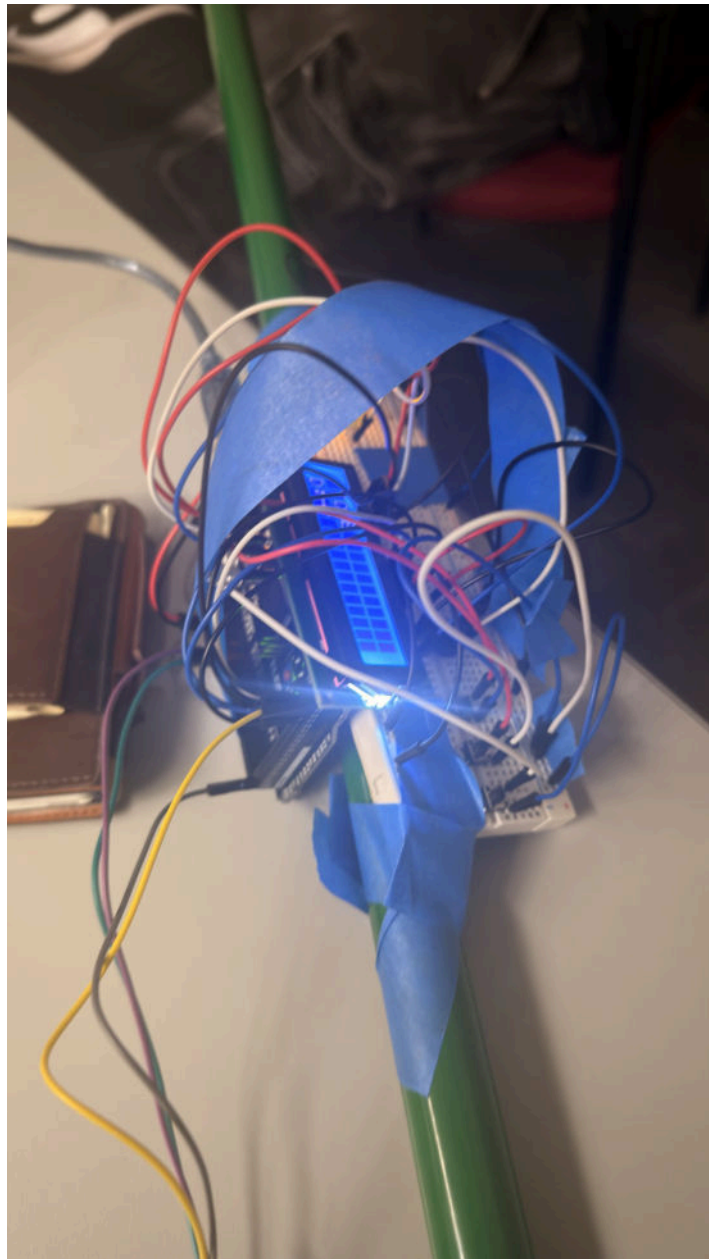
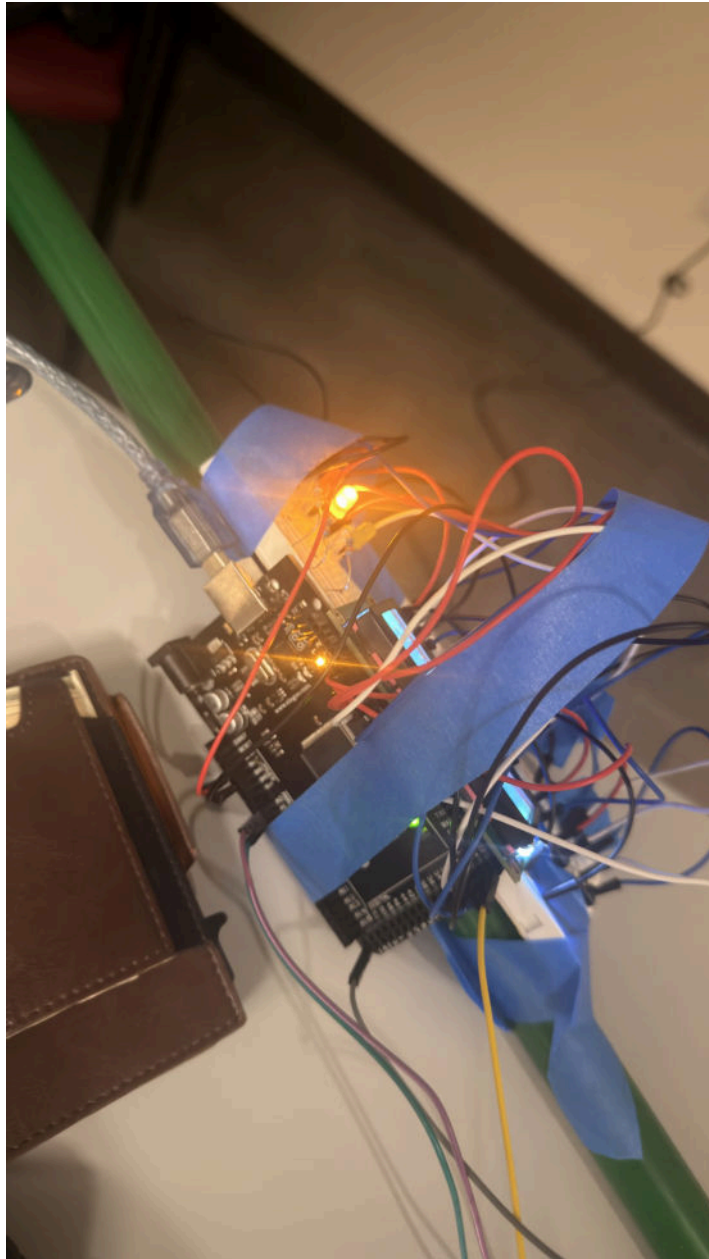


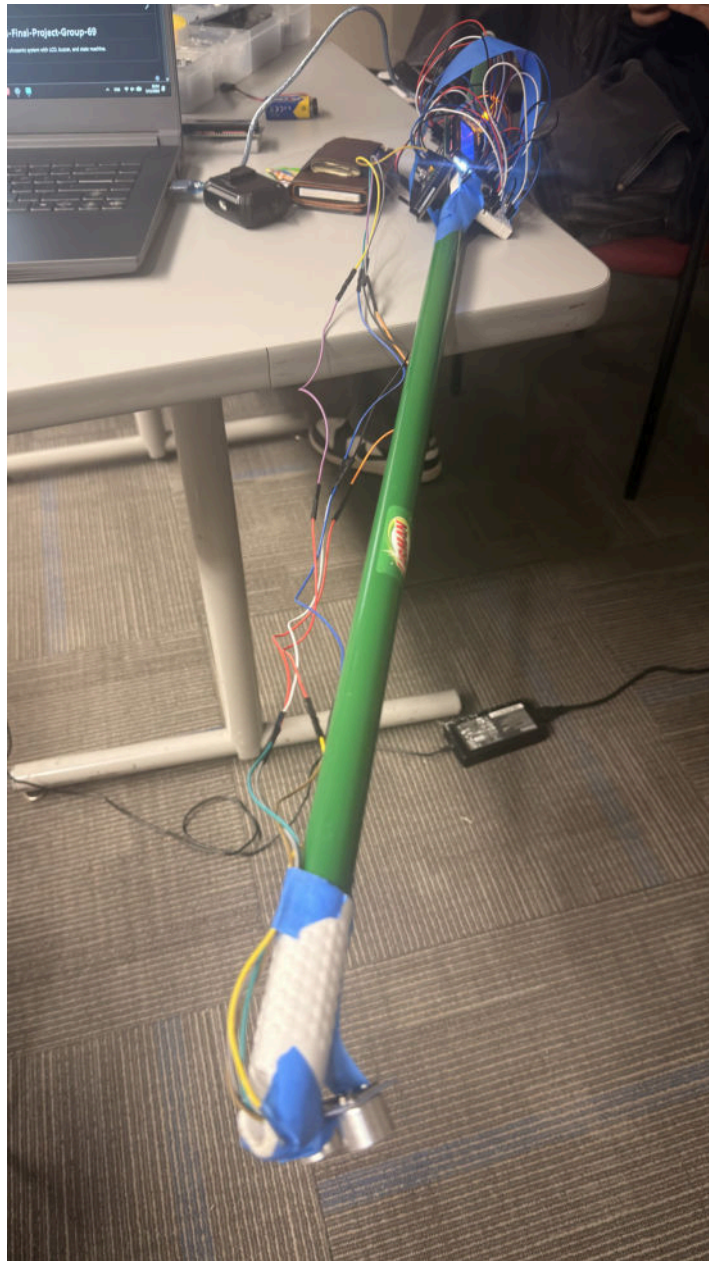
Figure 2: This is the flowchart, which depicts the system's four separate states, transition conditions, and each state's behavior

4 Result

Images of circuit:

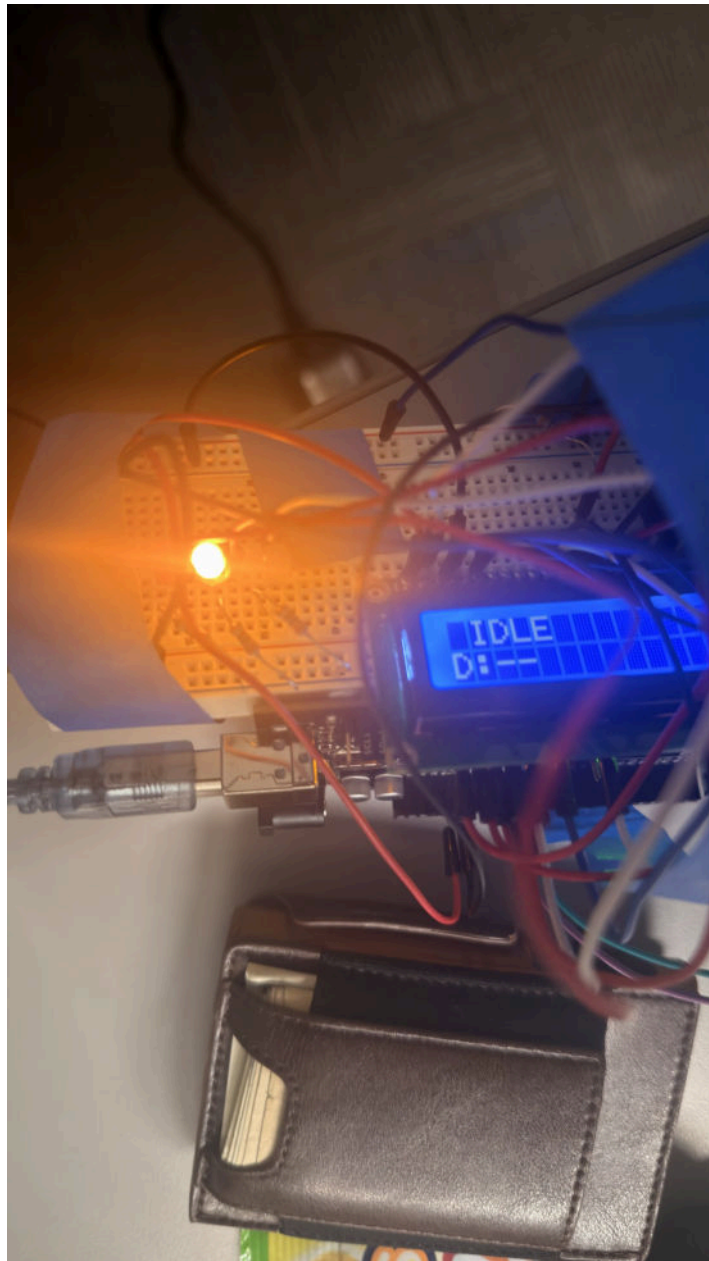






States: Off, Idle, Error, Active









Serial Monitor:

```
T:1 | OFF | D:0
T:2 | OFF | D:0
T:3 | OFF | D:0
T:4 | OFF | D:0
T:5 | IDLE | D:318
T:6 | IDLE | D:318
T:7 | IDLE | D:318
T:8 | ERROR | D:318
T:9 | ERROR | D:318
T:10 | ERROR | D:318
T:11 | ERROR | D:6
T:12 | ERROR | D:318
T:13 | ACTIVE | D:10
T:14 | IDLE | D:317
T:15 | IDLE | D:318
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5 Environmental Energy and Impact

Our project helps with reducing waste by having to only use one type of cane. Canes for blind people often come in different lengths and shapes for different situations, which means there is more production. In our case, only one type of cane is needed due to its versatility with different environments. It will be designed safely, as most of the circuit will be attached to the middle or top of the cane, only having the sensor at the bottom and a switch near the handle. These components are also cost effective, as it only uses one sensor for this to work. To house our circuits, we can use 3D printing with eco-friendly materials that are environmentally sustainable. The interface is also user-friendly, as there is only one switch that is being used to switch between two different modes.

6 Conclusion

Our system successfully works as intended. It correctly reads distance using the ultrasonic sensor, updates to one of the states (off, idle, active, error), and controls the LEDs, buzzer, and LCD display as intended. The ON, OFF, and RESET buttons also

function properly.

As for system limitations, there are a few. The ultrasonic sensor can occasionally give a wrong reading due to the angle it points at. Another limitation is the equipment we used, as we just taped our components to a pole.